

AI-Powered Regulatory Compliance Assistant

for EU Financial Institutions

RAG

Databricks

Azure OpenAI

FastAPI

LangGraph

GDPR

Meet the Team



Dimitris

Data Engineering & Databricks

- Delta Lake
Bronze/Silver/Gold
- PySpark ETL pipelines
- ADLS & Unity Catalog
- Workflow orchestration



Nontas

AI, RAG & Vector Search

- Azure OpenAI embeddings
- LangGraph orchestration
- Azure AI Search (hybrid)
- RAG evaluation pipelines



Panagiotis

API, DevOps & Governance

- FastAPI & REST endpoints
- CI/CD · Docker · GitHub
Actions
- PII controls & compliance

Business Context & Problem Statement

Business Context

Organizations face growing challenges managing large volumes of structured and unstructured enterprise knowledge across documents, databases, APIs, and cloud systems.

Traditional keyword-based search and manually maintained knowledge repositories are insufficient for modern AI-driven decision-making, semantic retrieval, and intelligent automation.

MiFID II

PSD2

DORA

GDPR

AML

EBA

The Problem

1. Compliance officers must search vast regulatory documents manually
2. Keyword search misses context and semantic relationships
3. Cross-document analysis is slow and error-prone
4. Regulatory risk increases without real-time, intelligent access

Project Objective & Solution Overview

Objective

Design and implement an AI-powered Regulatory Compliance Assistant enabling EU financial institutions to ingest, govern, semantically search, and analyze regulatory documentation using RAG, vector search, and enterprise AI orchestration — improving compliance efficiency and reducing regulatory risk.



Semantic Retrieval

RAG-powered natural language queries across EU regulatory corpus with citation-based answers



Governance & PII

PII detection, source traceability, lineage tracking, and responsible AI controls



Evaluation & Monitoring

Automated RAG evaluation, hallucination detection, and quality monitoring dashboards

Architecture — End-to-End Flow

Option 2

EU regulatory sources



EUR-Lex
Legal DB



EBA
Banking Auth.



ECB
Central Bank



ESMA
Markets Auth.



EU Comm.
Commission



Ingestion layer

Async pipelines · pagination · retry logic · incremental updates · document download



Data engineering — Databricks

Delta Lake Bronze/Silver/Gold · PySpark ETL · Unity Catalog · ADLS · orchestration workflows



Document processing

PDF/DOCX/HTML/XML parsing · chunking · metadata enrichment · deduplication · lineage tracking



AI processing — embeddings

Azure OpenAI · batch embedding pipelines · versioning · structured outputs · function calling



Retrieval + RAG engine

Azure AI Search · vector (HNSW) · hybrid BM25+cosine · LangGraph · citation grounding · hallucination detection



API & application layer — FastAPI

REST endpoints · JWT / OAuth2 · Swagger / OpenAPI · request validation · Azure Key Vault



Governance & security

PII detection · lineage · responsible AI · RBAC



DevOps & monitoring

GitHub Actions · Docker · CI/CD · eval dashboards

Technology Stack



Cloud & Data Platform

- Databricks
- Azure Data Lake Storage
- Delta Lake / Unity Catalog
- Azure OpenAI
- Azure AI Search
- **Azure Key Vault**



AI & Orchestration

- LangGraph
- OpenAI Embeddings
- RAG Pipelines
- **PySpark**
- Vector Stores
- Structured Outputs



Backend & DevOps

- FastAPI
- Python / uv
- Docker / Compose
- GitHub Actions
- Azure DevOps
- **JWT / OAuth2**



Data & Documents

- EUR-Lex API
- EBA / ECB APIs
- PDF / DOCX / HTML parsing
- Metadata enrichment
- PII detection
- Evaluation datasets

Dataset — EU Regulatory Documentation

Primary Sources

- European Banking Authority (EBA)
-
- European Securities & Markets Authority (ESMA)
-
- European Central Bank (ECB)
-
- EUR-Lex EU Legal Database
-
- DORA · PSD2 · MiFID II · AMLD
-
- GDPR · CRR/CRD · AI Act

Topics Covered

- KYC · Digital Finance · Risk Mgmt · Data Protection

Processing Pipeline

- 1 Text Extraction**
PDF / DOCX / HTML / JSON parsing
- 2 Chunking**
Semantic & fixed-size strategies
- 3 Metadata Enrichment**
Title, authority, date, category, domain
- 4 Deduplication**
Document lineage & provenance tracking
- 5 Embedding Generation**
Azure OpenAI embeddings → Delta Lake

RAG Architecture — End-to-End Flow



FastAPI endpoint

Azure OpenAI

Azure AI Search

LangGraph

GPT-4o

With citations

Vector Search

Semantic similarity retrieval using cosine distance on Azure AI Search index with HNSW algorithm

Hybrid Search

Combines BM25 keyword ranking with vector similarity for maximum recall across regulatory content

LangGraph Orchestration

Stateful multi-step AI workflows with conditional routing, retry logic, and citation extraction

DevOps, Infrastructure & CI/CD



Version Control & CI/CD

- Git branching strategy (feature / develop / main)
- Pull request & code review workflows
- GitHub Actions automated pipelines
- Azure DevOps integration



Containerization

- Docker containerization for all services
- Docker Compose multi-service environments
- Reproducible development & production setups
- Dependency isolation with uv



Secrets & Security

- Azure Key Vault for secrets management
- **JWT & OAuth2 authentication**
- Strong typing with Pydantic validation
- Environment separation (dev / staging / prod)



Monitoring & Quality

- RAG evaluation datasets & pipelines
- Hallucination detection mechanisms
- Automated quality assessment
- Monitoring dashboards & reporting metrics

Governance, PII & Compliance Controls



PII Detection & Redaction

Automated PII scanning and redaction on ingested regulatory documents before indexing and embedding generation.



Governance Metadata Models

Standardised metadata schema capturing regulation title, authority, publication date, category, version, and compliance domain.



Source Traceability & Lineage

End-to-end document lineage tracking from source acquisition through chunking, embedding, and retrieval to final AI response.



Responsible AI Controls

Citation-based answers enforce grounding; hallucination detection pipelines flag uncertain responses before surfacing to users.



Secure Data Handling

Azure Key Vault, encrypted ADLS storage, role-based access control, and JWT/OAuth2 enforced at API gateway level.

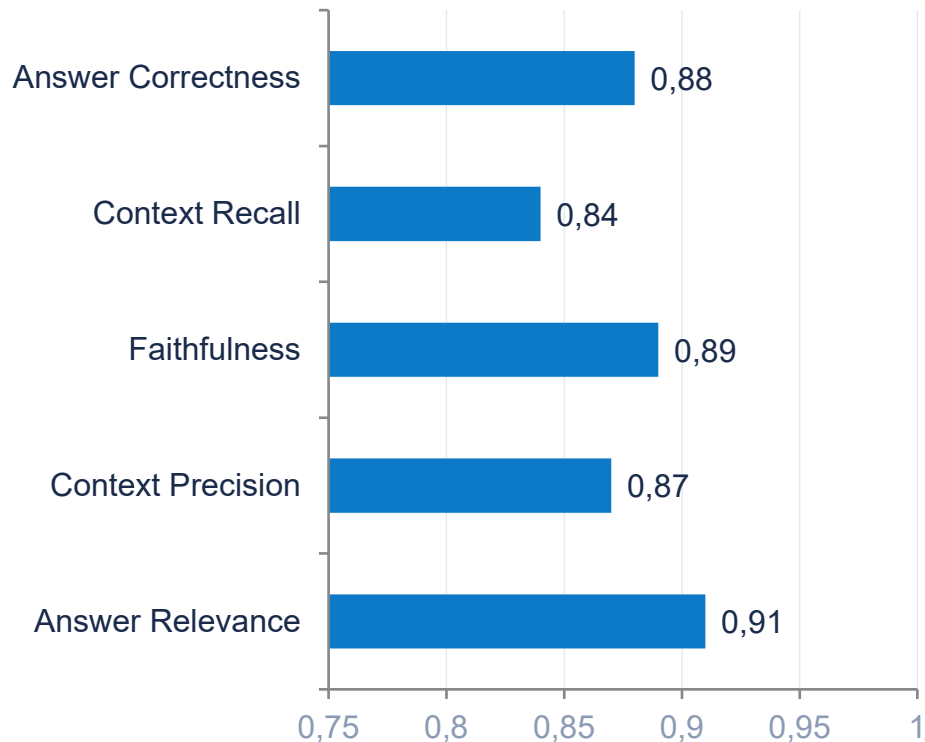


Audit & Compliance Logs

Immutable Delta Lake audit log capturing every ingestion event, transformation step, and user query with timestamps.

Evaluation Framework & Monitoring

RAG Evaluation Scores



Evaluation Datasets

Curated Q&A pairs covering MiFID II, GDPR, DORA, and AML compliance questions for automated scoring.

Hallucination Detection

Faithfulness scores computed by comparing generated answers against retrieved context chunks.

Automated Pipelines

Scheduled evaluation runs on every model update; results logged to Delta Lake evaluation tables.

Dashboards & Reports

Real-time quality dashboards exposing precision, recall, latency, and user feedback metrics.

Project Management Approach

01

Initialization

Repo setup · Python env · uv · typing · standards

02

Ingestion & Eng.

EUR-Lex/EBA/ECB pipelines · Databricks · Delta Lake

03

AI Processing

Document parsing · embeddings · metadata enrichment

04

Retrieval & RAG

Vector search · hybrid retrieval · LangGraph workflows

05

Governance & Eval

PII · compliance controls · evaluation · monitoring

Work tracked via shared task board · Regular sync meetings · Key decisions logged · One Project Coordinator per team

Evaluation Criteria



Technical Implementation Quality

Code quality, architectural patterns, use of best practices, strong typing, testing coverage, and production-readiness of all implemented components.



Architectural Correctness

Correct separation of concerns across data, AI, retrieval, API, and governance layers; adherence to the Bronze/Silver/Gold Delta Lake pattern.



Operational Reliability

CI/CD pipeline stability, Docker containerisation, retry/fault-tolerance mechanisms, secrets management, and monitoring instrumentation.



Business Relevance & Usability

The assistant meaningfully addresses real EU compliance use cases; natural language queries return grounded, cited, actionable answers.



Demonstration Completeness

End-to-end demo covering ingestion, retrieval, RAG response, evaluation results, and architectural overview presented clearly by all team members.

Thank You

AI-Powered Regulatory Compliance Assistant for EU Financial Institutions

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Questions & Demo